

Linear equations and algebraic systems

While TSM 099 studies Topic 34. Review: Topic 24, Topic 25, Topic 26, Topic 28, Topic 33.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: equations with parentheses

1. Solve and check in the original equation.

$$-2x + 10 = x + 22$$

$$x = -4$$

Check by substituting into the original equation.

2. Solve and check in the original equation.

$$6x + 9 = -3x$$

$$x = -1$$

Check by substituting into the original equation.

3. Solve and check in the original equation.

$$-2x + 2 = 3x + 2$$

$$x = 0$$

Check by substituting into the original equation.

4. Solve and check in the original equation.

$$3x + 13 = 2x + 14$$

$$x = 1$$

Check by substituting into the original equation.

5. Solve and check in the original equation.

$$6x - 8 = -26$$

$$x = -3$$

Check by substituting into the original equation.

6. Solve and check in the original equation.

$$-2x - 5 = -x - 4$$

$$x = -1$$

Check by substituting into the original equation.

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Core: systems by substitution

7. Solve by substitution; check both equations.

$$2x + (1)(3x - 11) = 9$$

$$(x, y) = (4, 1)$$

$$-12 + (1) = -11; \quad 8 + (1) = 9$$

Substitute the resulting x and y into both original equations.

8. Solve by substitution; check both equations.

$$4x + (1)(3x + 1) = 1$$

$$(x, y) = (0, 1)$$

$$0 + (1) = 1; \quad 0 + (1) = 1$$

Substitute the resulting x and y into both original equations.

9. Solve by substitution; check both equations.

$$2x + (2)(3x + 10) = 4$$

$$(x, y) = (-2, 4)$$

$$6 + (4) = 10; \quad -4 + (8) = 4$$

Substitute the resulting x and y into both original equations.

10. Solve by substitution; check both equations.

$$3x + (1)(-2x) = 0$$

$$(x, y) = (0, 0)$$

$$0 + (0) = 0; \quad 0 + (0) = 0$$

Substitute the resulting x and y into both original equations.

11. Solve by substitution; check both equations.

$$2x + (1)(-3x + 10) = 8$$

$$(x, y) = (2, 4)$$

$$6 + (4) = 10; \quad 4 + (4) = 8$$

Substitute the resulting x and y into both original equations.

12. Solve by substitution; check both equations.

$$4x + (2)(3x + 2) = 4$$

$$(x, y) = (0, 2)$$

$$0 + (2) = 2; \quad 0 + (4) = 4$$

Substitute the resulting x and y into both original equations.

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Core: systems by elimination

13. Solve by elimination; check both equations.

$$(1)E_1 + (1)E_2 : 4x = 16$$

$$(x, y) = (4, 1)$$

$$8 + (3) = 11; \quad 8 + (-3) = 5$$

Substitute the resulting x and y into both original equations.

14. Solve by elimination; check both equations.

$$(2)E_1 + (3)E_2 : 14x = 0$$

$$(x, y) = (0, 1)$$

$$0 + (3) = 3; \quad 0 + (-2) = -2$$

Substitute the resulting x and y into both original equations.

15. Solve by elimination; check both equations.

$$(1)E_1 + (1)E_2 : 6x = -12$$

$$(x, y) = (-2, 4)$$

$$-4 + (8) = 4; \quad -8 + (-8) = -16$$

Substitute the resulting x and y into both original equations.

16. Solve by elimination; check both equations.

$$(2)E_1 + (3)E_2 : 8x = 0$$

$$(x, y) = (0, 0)$$

$$0 + (0) = 0; \quad 0 + (0) = 0$$

Substitute the resulting x and y into both original equations.

17. Solve by elimination; check both equations.

$$(3)E_1 + (2)E_2 : 13x = 26$$

$$(x, y) = (2, 4)$$

$$6 + (8) = 14; \quad 4 + (-12) = -8$$

Substitute the resulting x and y into both original equations.

18. Solve by elimination; check both equations.

$$(3)E_1 + (2)E_2 : 11x = 0$$

$$(x, y) = (0, 2)$$

$$0 + (4) = 4; \quad 0 + (-6) = -6$$

Substitute the resulting x and y into both original equations.

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Extension: mixed equations and systems

19. Solve by elimination; check both equations.

$$(2)E_1 + (3)E_2 : 13x = 26$$

$$(x, y) = (2, 1)$$

$$4 + (3) = 7; \quad 6 + (-2) = 4$$

Substitute the resulting x and y into both original equations.

20. Solve by substitution; check both equations.

$$3x + (2)(2x + 5) = 8$$

$$(x, y) = \left(-\frac{2}{7}, \frac{31}{7}\right)$$

$$\frac{4}{7} + \left(\frac{31}{7}\right) = 5; \quad -\frac{6}{7} + \left(\frac{62}{7}\right) = 8$$

Substitute the resulting x and y into both original equations.

21. Solve by elimination; check both equations.

$$(2)E_1 + (-1)E_2 : 0x = 0$$

Infinitely many solutions; the second equation is a multiple of the first.

22. Solve by elimination; check both equations.

$$(2)E_1 + (-1)E_2 : 0x = -3$$

No solution; eliminating a variable produces a contradiction.

23. Solve and check in the original equation.

$$3x + 6 = 3x + 6$$

All real numbers (the two sides are identical).

24. Solve and check in the original equation.

$$4x - 5 = 2x + 3$$

$$x = 4$$

Check by substituting into the original equation.